



Transport across plasma membrane:

Type 2: Active transport

Dr. Niyan Jassim Mohammad

MBChB, MSc medical physiology

Department of medical physiology and microbiology

College of medicine

Hawler medical university

2025-2026

OBJECTIVE

- What is the key difference between passive and active processes?
- How do symporters and antiporters carry out their functions?
- What are the sources of cellular energy for active transport?
- In what ways are endocytosis and exocytosis similar and different?

Active Processes: Active Transport:

- Some polar or charged solutes that must enter or leave body cells cannot cross the plasma membrane by passive transport because they need to move against their concentration gradients. Such solutes cross the membrane by **active transport**.
- In active transport energy is required for carrier proteins to move solutes across the membrane against concentration gradient.

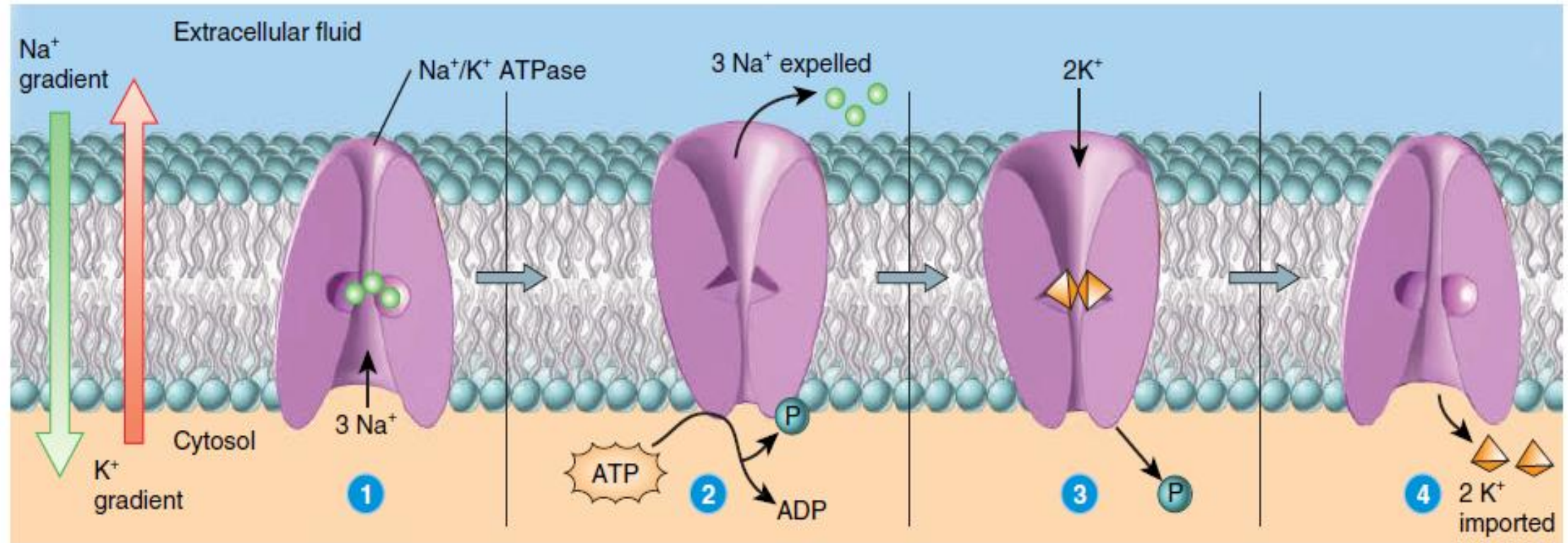
Primary Active Transport:

- In **primary active transport**, energy derived from hydrolysis of ATP changes the shape of a carrier protein, which “pumps” a substance across a plasma membrane against its concentration gradient.
- Carrier proteins that mediate primary active transport are often called **pumps**.
- The most prevalent primary active transport mechanism expels sodium ions (Na) from cells and brings potassium ions (K) in. Is called **sodium–potassium pump**, it also **Na–K ATPase**.

- All cells have thousands of **sodium–potassium pumps** in their plasma membranes. These pumps maintain a low concentration of Na in the cytosol by pumping Na into the extracellular fluid against the Na concentration gradient. At the same time, move K into cells against the K concentration gradient.
- K and Na slowly leak back across the plasma membrane down their electrochemical gradients—through passive transport or secondary active transport. So; the sodium–potassium pumps must work nonstop to maintain a low concentration of Na and a high concentration of K in the cytosol.

Figure 3.10 The sodium–potassium pump (Na^+/K^+ ATPase) expels sodium ions (Na^+) and brings potassium ions (K^+) into the cell.

 Sodium–potassium pumps maintain a low intracellular concentration of sodium ions.



3 sodium ions (Na^+) from the cytosol bind to the inside surface of the sodium–potassium pump.

Na^+ binding triggers ATP to bind to the pump and be split into ADP and P (phosphate). The energy from ATP splitting causes the protein to change shape, which moves the Na^+ to the outside.

2 potassium ions (K^+) land to the outside surface of the pump and cause the P to be released.

The release of the P causes the pump to return to its original shape, which moves the K^+ into the cell.

Secondary Active Transport

- In **secondary active transport**, the energy stored in a Na or H concentration gradient is used to drive other substances across the membrane against their own concentration gradients.
- Because a Na or H gradient is established by primary active transport, secondary active transport *indirectly* uses energy obtained from the hydrolysis of ATP.
- The sodium–potassium pump maintains a steep concentration gradient of Na across the plasma membrane. As a result, the sodium ions have stored or potential energy, just like water behind a dam. Accordingly, if there is a route for Na to leak back in, some of the stored energy can be converted to kinetic energy and used to transport other substances *against their concentration gradients*.

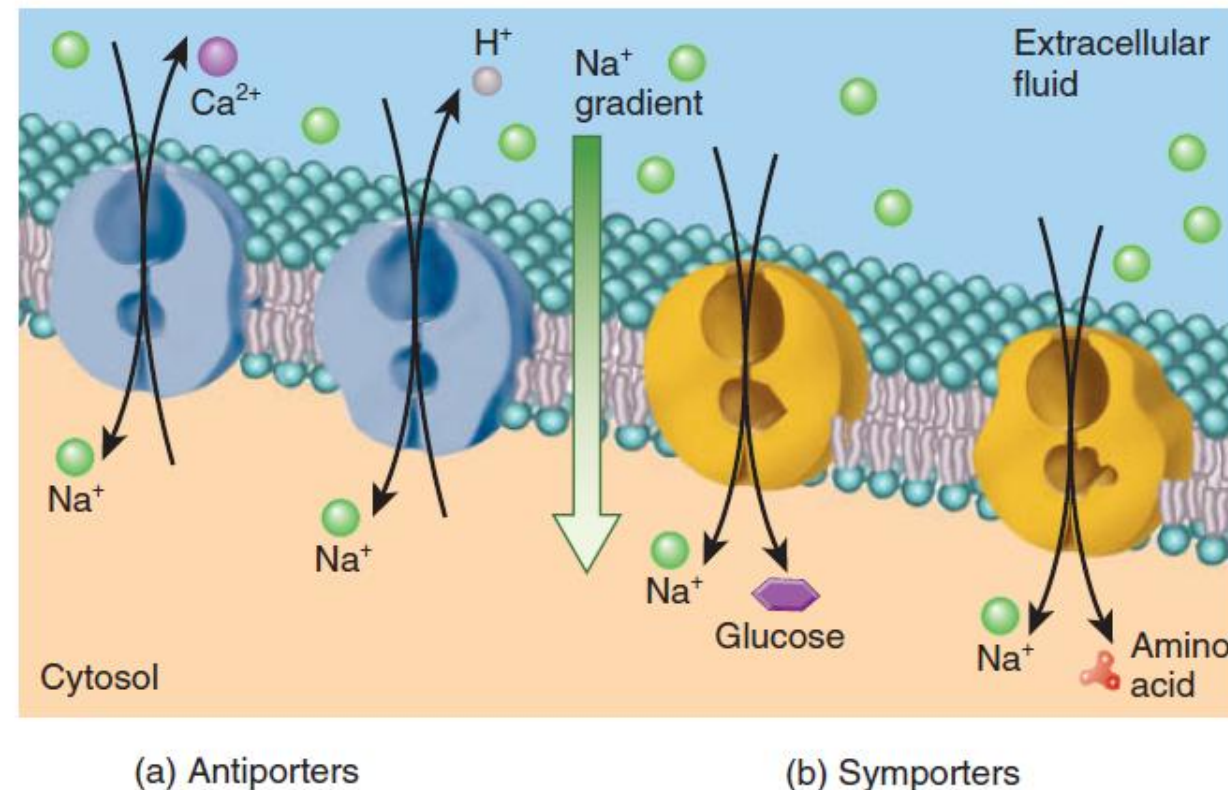
- Secondary active transport proteins harness the energy in the Na concentration gradient by providing routes for Na to leak into cells.
- In secondary active transport, a carrier protein simultaneously binds to Na and another substance and then changes its shape so that both substances cross the membrane at the same time.
- If these transporters move two substances in the same direction they are called **symporters**, in contrast, **antiporters** move two substances in opposite directions across the membrane.
- Plasma membranes contain several antiporters and symporters that are powered by the Na gradient, for instance, the concentration of calcium ions is low in the cytosol because Na–Ca antiporters eject calcium ions.

- Likewise, Na–H antiporters help regulate the cytosol's pH (H concentration) by expelling excess H.
- By contrast, dietary glucose and amino acids are absorbed into cells that line the small intestine by Na–glucose and Na–amino acid symporters. In each case, sodium ions are moving down their concentration gradient while the other solutes move “uphill,” against their concentration gradients.
- All these symporters and antiporters can do their job because the sodium–potassium pumps maintain a low concentration of Na in the cytosol.

Figure 3.11 Secondary active transport mechanisms.

(a) Antiporters carry two substances across the membrane in opposite directions. (b) Symporters carry two substances across the membrane in the same direction.

 Secondary active transport mechanisms use the energy stored in an ionic concentration gradient (here, for Na^+). Because primary active transport pumps that hydrolyze ATP maintain the gradient, secondary active transport mechanisms consume ATP indirectly.



Transport in Vesicles

- A **vesicle** is a small, spherical sac.
- A variety of substances are transported in vesicles within cells. Vesicles also import materials from and release materials into extracellular fluid.
- During **endocytosis**, materials move into a cell in a vesicle formed from the plasma membrane.
- In **exocytosis**, materials move out of a cell by the fusion with the plasma membrane of vesicles formed inside the cell.
- Both endocytosis and exocytosis require energy supplied by ATP.

Endocytosis:

Three types of endocytosis:

- 1) receptor-mediated endocytosis.
- 2) phagocytosis.
- 3) bulk-phase endocytosis.

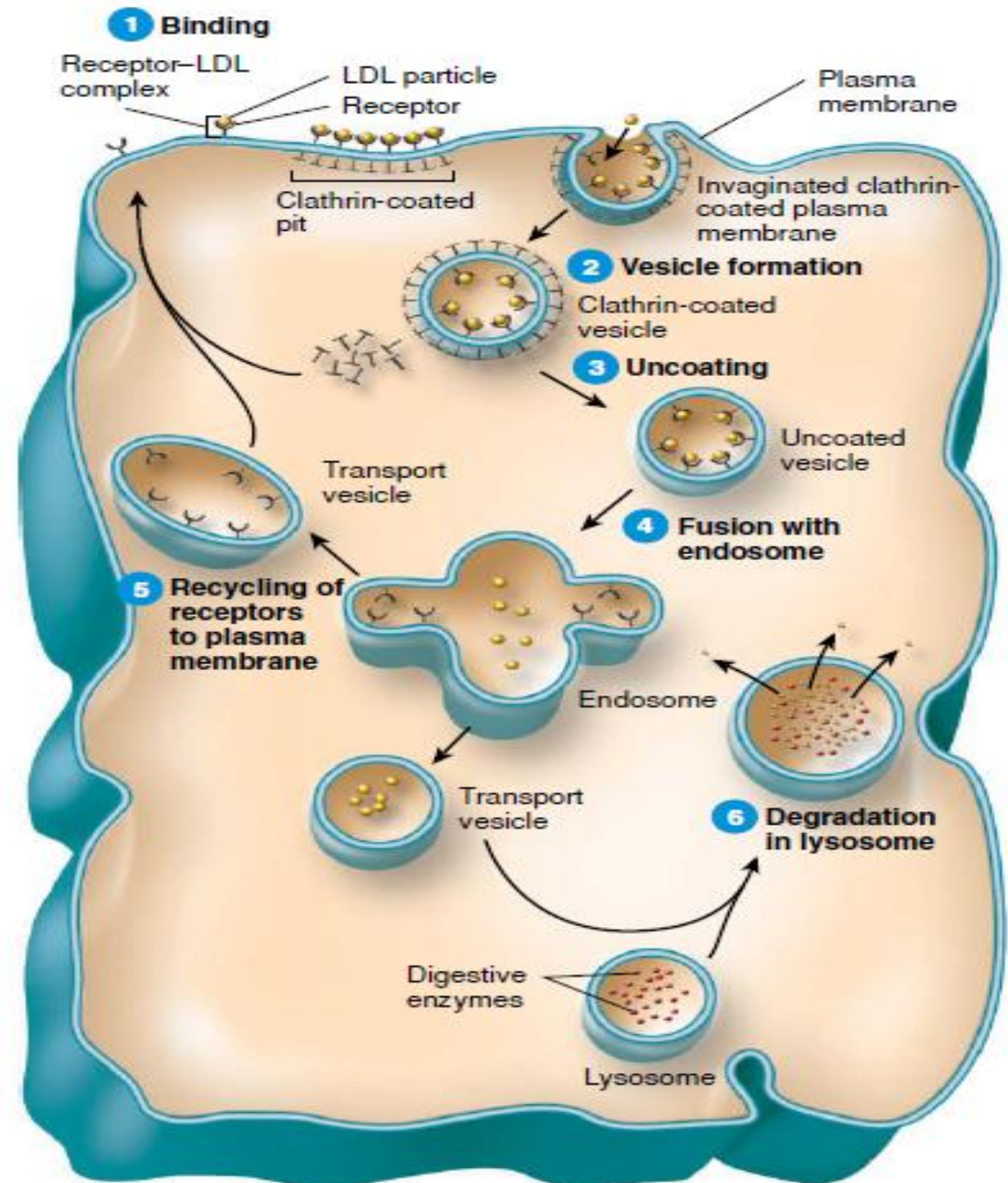
1. Receptor-mediated endocytosis

- Is a highly selective type of endocytosis by which cells take up specific ligands. (ligands are molecules that bind to specific receptors).
- A vesicle forms after a receptor protein in the plasma membrane recognizes and binds to a particular particle in the extracellular fluid. For instance, cells take up cholesterol-containing LDLs, transferrin (an iron transporting protein in the blood), some vitamins, antibodies, and certain hormones by receptor-mediated endocytosis.

Figure 3.12 Receptor-mediated endocytosis of a low-density lipoprotein (LDL) particle.



1. Binding: LDL particle that contains cholesterol binds to a specific receptor in the plasma membrane to form a receptor–LDL complex. The receptors are concentrated in regions of the plasma membrane called clathrin-coated pits. Here, a protein called clathrin attaches to the membrane on its cytoplasmic side. Many clathrin molecules come together, forming a basketlike structure around the receptor–LDL complexes that causes the membrane to invaginate.



2. Vesicle formation. The invaginated edges of the membrane around the clathrin-coated pit fuse, and a small piece of the membrane pinches off. The resulting vesicle, known as a clathrin-coated vesicle, contains the receptor–LDL complexes.

3. Uncoating. Immediately after it is formed, the clathrin-coated vesicle loses its clathrin coat to become an uncoated vesicle. Clathrin molecules either return to the inner surface of the plasma membrane or help form coats on other vesicles inside the cell.

4. Fusion with endosome. The uncoated vesicle quickly fuses with a vesicle known as an endosome. Within an endosome, the LDL particles separate from their receptors.

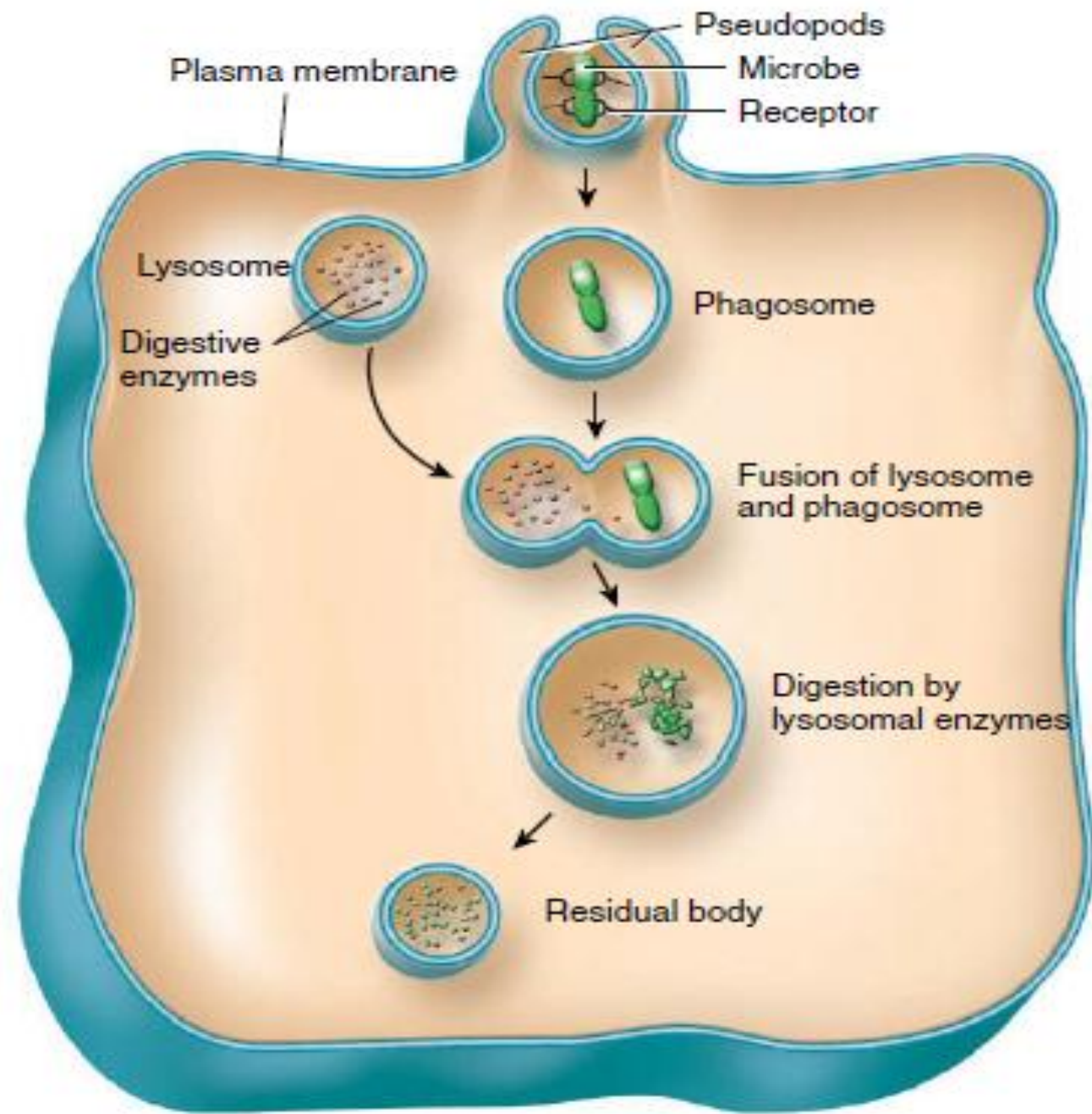
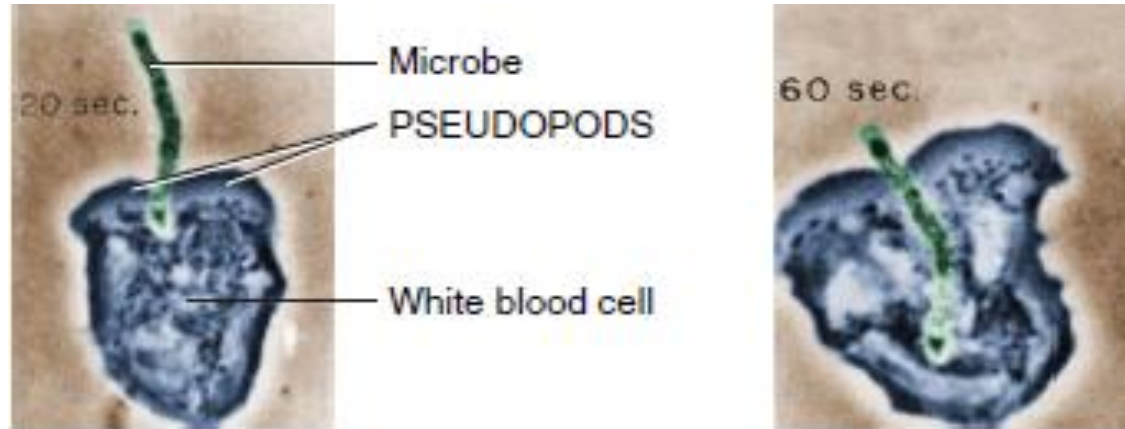
5. Recycling of receptors to plasma membrane. Most of the receptors accumulate in elongated protrusions of the endosome. These pinch off, forming transport vesicles that return the receptors to the plasma membrane. An LDL receptor is returned to the plasma membrane about 10 minutes after it enters a cell.

6. Degradation in lysosomes. Other transport vesicles, which contain the LDL particles, bud off the endosome and soon fuse with a lysosome. Lysosomes contain many digestive enzymes. Certain enzymes break down the large protein and lipid molecules of the LDL particle into amino acids, fatty acids, and cholesterol. These smaller molecules then leave the lysosome. The cell uses cholesterol for rebuilding its membranes and for synthesis of steroids, such as estrogen. Fatty acids and amino acids can be used for ATP production or to build other molecules needed by the cell.

2. Phagocytosis:

- Is a form of endocytosis in which the cell engulfs large solid particles, such as worn-out cells, whole bacteria, or viruses.
- Only a few body cells, termed **phagocytes**, are able to carry out phagocytosis. Two main types of phagocytes are macrophages, located in many body tissues, and neutrophils, a type of white blood cell.
- Phagocytosis begins when the particle binds to a plasma membrane receptor on the phagocyte, causing it to extend **pseudopods**, projections of its plasma membrane and cytoplasm. Pseudopods surround the particle outside the cell, and the membranes fuse to form a vesicle called a phagosome, which enters the cytoplasm.
- The phagosome fuses with one or more lysosomes, and lysosomal enzymes break down the ingested material.
- In most cases, any undigested materials in the phagosome remain indefinitely in a vesicle called a residual body. The residual bodies are then either secreted by the cell via exocytosis or they remain stored in the cell as lipofuscin granules.

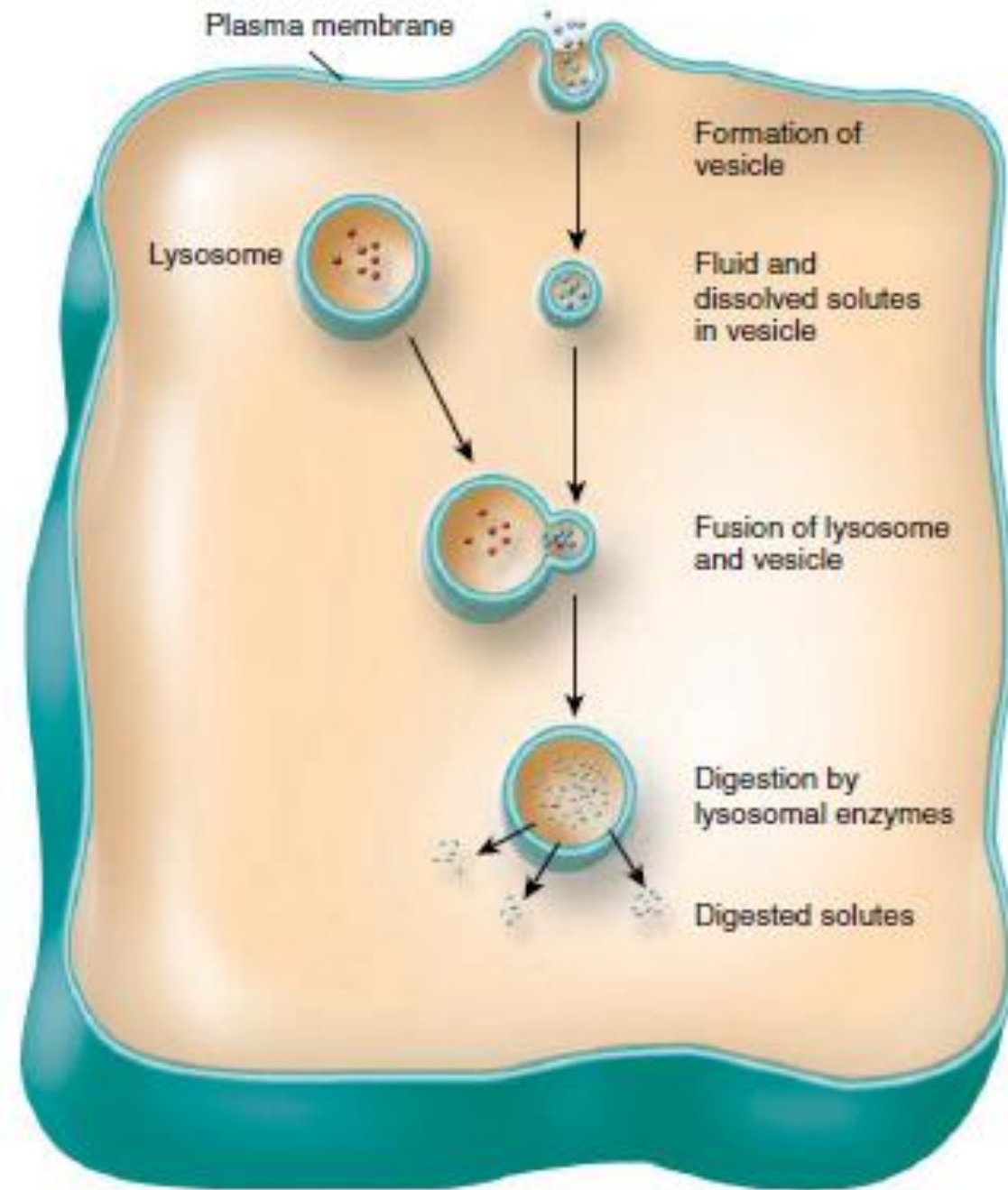
Figure 3.13 Phagocytosis. Pseudopods surround a particle, and the membranes fuse to form a phagosome.



(a) Diagram of the process

3. Bulk phase endocytosis:

- Also called **pinocytosis**.
- Tiny droplets of extracellular fluid are taken up.
- No receptor proteins are involved.
- Solutes dissolved in the extracellular fluid are brought into the cell.
- Plasma membrane folds inward and forms a vesicle containing a droplet of extracellular fluid.
- The vesicle detaches or “pinches off” from the plasma membrane and enters the cytosol.
- Within the cell, the vesicle fuses with a lysosome, where enzymes degrade the engulfed solutes.
- The resulting smaller molecules, such as amino acids and fatty acids, leave the lysosome to be used elsewhere in the cell.
- It occurs in most cells, especially absorptive cells in the intestines and kidneys.



Exocytosis:

- Exocytosis releases materials from a cell.
- All cells carry out exocytosis, but it is especially important in two types of cells:
 - (1) secretory cells that liberate digestive enzymes, hormones, mucus, or other secretions.
 - (2) nerve cells that release substances called *neurotransmitters*.
- In some cases, wastes are also released by exocytosis. During exocytosis, membrane-enclosed vesicles called *secretory vesicles* form inside the cell, fuse with the plasma membrane, and release their contents into the extracellular fluid.



• **THANK YOU** •

■
ANY QUESTION?